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ORIGINAL ARTICLE



Entropy method of constructing a combined model for improving loan default prediction: A case study in China

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ABSTRACT

In recent years, credit scoring has become an efficient tool to assist financial institutions in identifying potential default borrowers, and the combined model is widely viewed as a useful vehicle. In this study, after pre-processing based on random forest, we propose a combined logistic regression algorithm and artificial neural network model to improve the predictive performance based on actual data from a rural commercial bank under the condition that loan quality directly affects the profitability of the bank. The combined model requires a step with an entropy method to determine the entropy weights of the logistic regression model and artificial neural network model. The experimental results reveal that the proposed combined model outperforms the two base models on four evaluation metrics: accuracy (ACC), area under the curve (AUC), Kolmogorov-Smirnov statistic (KS), and Brier score (BS). Moreover, the model is superior to a state-of-the-art ensemble model, stacking.

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Credit scoring; combined model; entropy method; stacking

1. Introduction

1.1. Background

Credit loan quality directly affects the economic benefit of commercial banks in China; therefore, it is meaningful for banks to avoid unnecessary loss by developing credit scoring models. Banks are now cognisant of the need to adopt rigorous credit evaluation models in their systems when reviewing a loan to an individual client or company. On May 15th, 2019, the People's Bank of China introduced a lower preferential reserve ratio for small- and medium-sized banks serving local regions. Approximately 1,000 county-level rural commercial banks can enjoy the preferential policy, freeing up approximately 280 billion (yuan) of long-term funds, all of which will be used to extend loans. Most of the groups served by rural commercial banks are peasants or rural people, and collecting information from these people is not as easy and comprehensive as collecting information from people in cities. Meanwhile, with the rapid increment of gross domestic product (GDP), new loans have increased by more than 16 trillion (yuan) in 2018 in China. As the manual estimation of customer creditworthiness in rural commercial banks has become both time and resource consuming (Ala'raj & Abbod, 2016a), some customers of rural commercial banks are being attracted by other financial institutions and Internet financial companies, which has led to the development and

implementation of programming models for loan estimation to improve the efficiency of the loan process. Therefore, rural commercial banks urgently need to focus on credit score modelling to improve their risk control and capital allocation abilities.

Generally, credit scoring models are used to estimate and predict a new applicant's credit risk or probability of default on the basis of past loan applicants' information. Evaluating loan quality and controlling loan default risk have attracted worldwide attention of scholars. Credit scoring methods proposed by scholars can be divided into linear discriminant analysis, regression analysis, non-parametric, artificial intelligence (Zhu, Xie, Sun, Wang, & Yan, 2016) and quantification methods. Eisenbeis (Eisenbeis, 1977) proposed a discriminant analysis scoring method for the risk assessment of consumer credit and showed that the method focused on only the problem of minimizing the default rate and lacked consideration of maximizing the profit of borrowers. Orgler (Orgler, 1970) applied linear regression to model commercial loans and designed a score-card to evaluate unpaid loans. As the unpaid loans contained behavioural information about the borrowers, he found that the behavioural characteristics of borrowers were more indicative of loans' future quality than were the characteristics of applicants. Amemiya (Amemiya, 1981) conducted an empirical analysis of urban consumer credit data and concluded that logistic regression had strong robustness. Grablowsky and Talley (Grablowsky

& Talley, 1981) used both the discriminant model and probit regression in credit scoring research and found that the results of the latter were better than those of the former when the outputs of both models were the probability of repayment or default. Subsequently, non-parametric methods, such as classification trees, k-nearest neighbour discrimination and clustering analysis, have gradually been applied in personal credit scoring models to assess credit risks (Granger & Ramanathan, 1984). Makowski (Makowski, 1985) and Coffman (Coffman, 1986) applied the classification tree method to credit evaluation. Henley and Hand (Henley & Hand, 1996) conducted deep research on the application of the k-nearest neighbour method in credit risk assessment. Desai, Crook and Overstreet (Desai, Crook, & Overstreet Jr, 1996) compared the neural network model with the linear scoring model in the context of credit cooperatives and found that the neural network model achieved good prediction accuracy in terms of the correct classification rate of bad debts. Yonghan Ju and So Young Sohn proposed a credit-risk rating model based on the quantification score (Ju & Sohn, 2017).

Chinese scholars have made progress in the research of credit scoring models. Cui Yuanyuan (Yuanyuan, 2006) used logistic regression, neural networks and classification trees to model and analyse the auto loan data of a group of commercial banks. Shu Yang (Yang, 2017) proposed that the logistic model was applicable to the prediction of customer default; Shen Cuihua (Cuihua, 2004) used a support vector machine (SVM) to evaluate personal loan applications and obtained better results than those of the k-nearest neighbour method. Jiang Minghui (Minghui, 2007) proposed using the posterior probability of the SVM model to overcome the deficiency of dividing customers into default or non-default customers for housing credit evaluation in the evaluation of the SVM model. Shi Qingyan (Qingyan, 2005) proposed a two-stage mixed personal credit scoring model based on a neural network and logistic regression that achieved good accuracy and robustness. Li Zhengbo (Zhengbo, 2006) empirically researched the loan credit risk of farmers in rural credit cooperatives based on survey data and found that the impact factors of peasant default behaviour included planting income, proprietary spending, loan term, interest rate and rural credit cooperatives' service. In addition, the age and education of farmers had a certain influence. Yu Jiang (Jiang, 2007) and Chen Donghai (Donghai, 2005) showed that the personal characteristics of borrowers, guarantees, loan interest rates, social credit environment and the bank's ability to control credit risk were indicators of whether lenders experienced moral hazard.

Many individual models, such as statistical methods, artificial intelligence techniques (e.g. neural networks, SVMs and random forests), and machine learning methods, have been applied to the classification of loan defaults (Saia & Carta, 2017, 2018), but each individual model has its own disadvantages and may contribute a certain number of defaults to the final prediction inaccuracy. Meanwhile, such techniques can be combined to develop new hybrid approaches of credit scoring (Saia & Carta, 2016a). Therefore, researchers have experimented with combinations of two or more models to solve the credit classification problem. The classifier combination techniques can be divided into homogenous and heterogeneous classifier ensembles (Ala'raj & Abbod, 2016a), where the former combines classifiers of the same type, while the latter combines those of different types. The focus of this paper is the second group of techniques. This paper proposes a heterogeneous credit scoring model based on actual data from a rural commercial bank in China to supplement the research on loan default in rural commercial banks in China and to provide suggestions to control loan risk. The creation of this complex model is the main aim of this paper.

1.2. Research motivation

The main research motivation in this paper focuses on two aspects: 1) special characteristics of the raw data and 2) the LNN-Entropy combination credit scoring model. The first motivation of this paper is to consider the special characteristics of the raw data in the design process of the loan evaluation model. The characteristics that differentiate our raw data from those of other studies (Ala'raj & Abbod, 2016a; Xia, Liu, Da, & Xie, 2018; Xia, Liu, Li, & Liu, 2017; Xia, Liu, & Liu, 2017) are the following. 1) Rural commercial banks serve mostly rural residents, and traditional peasants in China are lawful and rarely default on loans, which makes the default loan rate lower than that of other types of financial institutions. 2) This rural commercial bank does not have its own loan risk control system, so it needs the lender to fill in some information when reviewing the loan, which results in fewer characteristic variables than expected in our initial data. 3) The overall data volume is large; thus, constructing a new credit scoring model to improve the prediction accuracy is challenging. One solution is to perform feature selection on the original data to remove the redundant variables and select the factors that have a substantial effect on the dependent variable. The other is to reduce the amount of the original data by filtering or removal (Ala'raj & Abbod, 2016a).

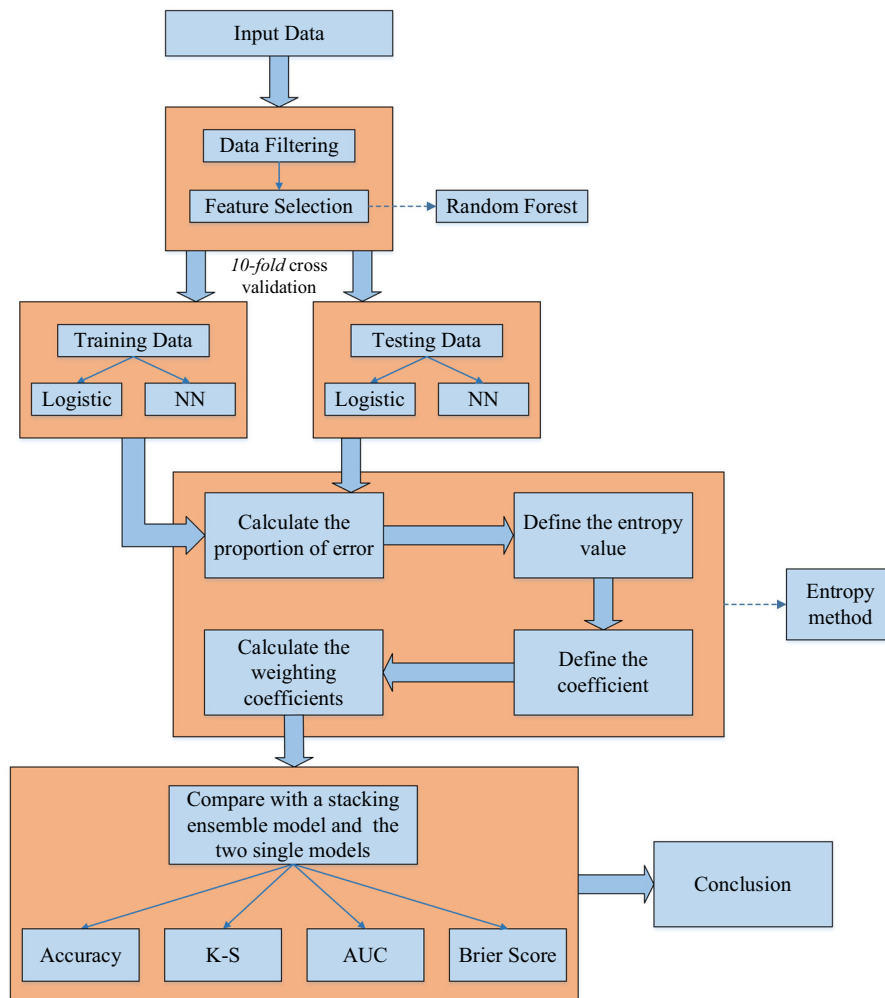


Figure 1. Flowchart of the heterogeneous combined credit scoring model.

The second motivation is to enhance the performance of the credit scoring model by applying the entropy method. The current research uses ensemble classifiers rather than individual classifiers to solve credit scoring problems. According to Lessmann, ensemble classifiers can be divided into two types (Lessmann, Seowb, Baenscd, & Thomasd, 2013): homogenous ensembles, which create base models using a single type of classification algorithm, and heterogeneous ensembles, which employ different types of classification algorithms. As suggested in (Santana, Silva, Canuto, Pintro, & Vale, 2010), when using heterogeneous ensembles, the good performance of one classifier can be used to overcome the poor performance of another individual classifier. In parallel with the growth in credit scale in the financial sector, linear discriminant analysis (LDA) and logistic regression analysis (LRA) have been widely used in credit scoring (Chuang & Lin, 2009; Crook, Edelman, & Thomas, 2007; Lee, Chiu, Chou, & Lu, 2006; Thomas, 2000; West, 2000). However, LDA is criticized for the assumption of a linear relationship between the dependent and independent variables, which is sensitive to deviations from the multivariate normality

assumption (Karels & Prakash, 1987; Reichert, Cho, & Wagner, 1983). In contrast to LDA, LRA does not require the normality assumption. Artificial neural network (ANNs) are another important alternative in financial prediction studies and has been comprehensively used in developing credit scoring models as an alternative to traditional statistical approaches such as LRA (Ala'raj & Abbod, 2015). The advantages of ANN models are the ability to handle incomplete, missing and noisy data and the fact that no assumptions about the distributions of the inputs are required. Additionally, ANNs can recognize complex patterns between variables (Ala'raj & Abbod, 2015). In our research, we choose LRA and ANN, in addition to heterogeneous methods, as the base models because they are both popular and have undergone experimental testing in numerous studies. The advantage of heterogeneous credit scoring models is that each technology contributes its own strengths.

In addition, Abbod (Ala'raj & Abbod, 2016a, 2016b) introduced the random forest in their heterogeneous ensemble model, which inspired us to include other individual classifiers in the heterogeneous ensemble model to enhance the predictive

capability of our model. Wang et al. (Wang, Hao, Ma, & Jiang, 2011) examined the performance of three popular ensemble methods (bagging, boosting, stacking) for credit scoring problems in terms of the average accuracy, type I error and type II error across three datasets and found that stacking and bagging decision tree achieved the best results. For a training set, the expected error of a classifier can be divided into three terms: variance, bias and noise (Bauer & Kohavi, 1999). Variance measures the variability of the predictions over changes in the training set; bias measures the proximity of the prediction to the target; and noise measures the irreducible part of the error that is due the variability in the data. Meanwhile, stacking mainly attempts to produce strong models that are less biased than the components; even the variance can also be reduced. Based on the two points above, we considered one of the popular ensemble methods, stacking, as the contrast to validate the effectiveness of the proposed novel combined model.

1.3. Contribution

On the basis of the above motivations, we propose a novel credit scoring model that combines LRA and ANN via the entropy method, in addition to data filtering and a feature selection algorithm. The contribution of this study is twofold. First, this study proposes a novel credit scoring model (LNN-Entropy model), and the empirical results validate this new model via comparison with a state-of-the-art approach, i.e. stacking. Second, to the best of our knowledge, only a few scholars (Saia & Carta, 2016b) have studied the use of the entropy method in the field of credit scoring; consequently, we apply the entropy method to the loan evaluation problem in the bank lending context and supplement the research field.

The proposed model is composed of the following phases (Figure 1): 1) collecting the dataset; 2) data filtering and feature selection; 3) splitting the data in training and testing sets; 4) generating the classification algorithms; 5) combining the two classifiers into an ensemble; and 6) assessing the model on the basis of performance evaluation metrics for two single models and a state-of-the-art ensemble model.

The rest of this paper is organized as follows: Section 2 introduces the methodologies applied in the empirical study and presents the random forest algorithm, logistic model, and BP neural network in detail. The interpretations of the parameters of empirical evidence are demonstrated in Section 3. Section 4 discusses the model performance, and Section 5 concludes this paper.

2. Research methodology

2.1. Random Forest algorithm

A random forest is a combination of decision trees used as an ensemble learning method for classification, regression and other tasks that operates by constructing a multitude of decision trees at training time and outputting the class that is the mode of the classification or regression of the individual trees (Kam, 1995). The algorithm was proposed by Leo Breiman (Breiman, 2001) for classification and prediction (Yaman, Subasi, & Rattay, 2018). A random forest is a type of learning method that can be used in classification and regression problems. Moreover, random forests are suitable for large quantities of data with substantial noise, can prevent overfitting, and are able to distinguish important features in classification (M. Deng, Chen, Huang, & Niu, 2018).

2.2. Logistic regression analysis (LRA)

Logistic regression is a widely applied statistical model that is a suitable approach for binary classification problems (Kim, Kim, Cho, Kim, & Hyun, 2018; Siddique, Akhtar, Lee, Kim, & Kim, 2017). Loans can be classified as default and no default, which matches the distribution of the logistic model. The formula for multivariate logistic regression is given as

$$\ln\left(\frac{P_i}{1 - P_i}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \mu_n \quad (1)$$

where P_i is the probability of default, $1 - P_i$ is the probability of no default, β_n is the estimated coefficient, X_n represents the independent variables and μ_n is the random error.

2.3. Artificial neural network (ANN)

The ANN is one of the most well-known algorithms for classification. In this study, a backpropagation neural network (BPNN) is utilized to train the ANN. BPNN is a non-linear feedforward network in which data flow forward and the error is propagated back to each layer of the network to adjust the model parameters, such as weights (Tian, You, & Huang, 2018). With a sufficient number of hidden layers and hidden layer neurons, BPNN can represent any non-linear mapping relationship and obtain optimal solutions. Therefore, in this paper, the BPNN is utilized to train the ANN.

2.4. LNN-Entropy model: The heterogeneous combined credit scoring model

In this paper, an entropy method-based credit scoring combined model is proposed. Entropy is a

Input: Dataset $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\}$;
 First-level learning algorithm $L_1, \dots, L_T$;
 Second-level learning algorithm L ;

Process:
 For $t = 1, 2, \dots, T$:
 $h_t = L_t(D)$ % Training a first-level individual learner h_t by applying the first-level
 end;
 % Learning an algorithm L_t to the original dataset D
 $D' = \emptyset$; % Generate a new dataset
 For $t = 1, 2, \dots, m$:
 For $t = 1, 2, \dots, T$:
 $z_{it} = h_t(x_i)$ % Use h_t to classify the training example x_i
 end;
 $D' = D' \cup \{(z_{i1}, z_{i2}, \dots, z_{iT}), y_i\}$
 end;
 $h' = L(D')$ % Training the second-level learner h' by applying the second-level
 % Learning algorithm L to the new dataset D'

Output: $H(x) = h'(h_1(x), h_2(x), \dots, h_T(x))$

Figure 2. The process of the stacking algorithm.

measure of uncertainty in information formulated using probability theory (H. Deng, Yeh, & Willis, 2000). The entropy method is an objective weighting method that uses the size of the information provided by the entropy values of each indicator or model to determine the weight of the indicator or model.

Suppose y_{it} is the predicted value of the t th variable by the i th evaluation method, where i ($i = 1, 2, \dots, m$) represents different evaluation methods and t ($t = 1, 2, \dots, n$) is an independent variable. y_t represents the actual value, and $e_{it} = y_t - y_{it}$ is the prediction deviation of the t th variable by the i th evaluation method. The entropy method includes the following 4 steps (H. Deng et al., 2000; Vatansever & Akgül, 2018):

Step 1: Calculating the proportion of error, converting e_{it} to p_{it} .

$$p_{it} = \frac{e_{it}}{\sum_{i=1}^n e_{it}} \quad (2)$$

Step 2: Defining the entropy value of the prediction error of the i th evaluation method as H_i . $H_i = -k \sum_{i=1}^n p_{it} \ln p_{it}$, where $k = \frac{1}{\ln n}$ is a constant that guarantees

$$0 \leq H_i \leq 1. \quad (3)$$

Step 3: Defining the coefficient

$$d_i = 1 - H_i, \quad 0 \leq d_i \leq 1. \quad (4)$$

Step 4: Calculating the weighting coefficients $w_1, w_2, \dots, w_m$ of the evaluation methods:

$$w_i = \frac{1}{m-1} \left(1 - \frac{d_i}{\sum_{i=1}^m d_i} \right), \quad \text{where } \sum_{i=1}^m w_i = 1. \quad (5)$$

2.5. Stacking

Stacking is a popular ensemble learning technique that combines multiple classifiers or regression

models via a meta-classifier or meta-regressor to achieve better predictive accuracy (Wolpert, 1992). The base-level models use the complete training set for training; then, the meta-model is trained on the outputs of the base-level model as features. The base-level model often consists of different learning algorithms; therefore, stacking ensembles are often heterogeneous. Figure 2 presents the pseudocode of the stacking algorithm (Wang et al., 2011).

3. Experimental set-up

3.1. Real-world credit dataset

The initial dataset consists of a set of loans from 2002 to 2016 provided by a rural commercial bank in China. The 35860 valid samples include 31595 no defaults and 4265 defaults. k -fold cross-validation is employed to enhance the model robustness and reduce the overfitting problem, with $k = 10$. Each subset is randomly shuffled and dataset is divided into k subsets, where a single subset is retained as the test set to evaluate the model, while the remaining $k-1$ subsets are used to train the model. The results of this process are averaged to generate the final result. To guarantee the reproducibility of the dividing process, the Python function *random.seed()* was implemented with the scikit-learn package in Python 3.6.

3.2. Data filtering

Loan evaluation indicators are generally divided into three types by both domestic and foreign commercial banks: credit analysis focused on borrowers' willingness to repay, financial analysis focused on borrowers' ability to repay, and comprehensive analysis of the above two situations. An increasing number of commercial banks in China use the

Table 1. Domestic commercial bank credit evaluation index system.

Banks				
Indicators		CCB	CMBC	HXB
Personal indicators	Age	+	+	+
	Gender	+		+
	Marital status	+		+
	Health condition	+		+
	Educational level	+	+	
	Household registration	+	+	+
	Type of work	+	+	+
	Work situation	+		
	Work position	+	+	+
	Current work duration	+	+	+
	Title	+		+
Economic indicators	Personal income	+	+	+
	Average monthly household income	+		+
	Net household income		+	
	Monetary assets		+	+
	Other assets		+	+
	Insurance		+	
Credit indicators	Have a bank account or not	+	+	+
	Reserve balance	+		
	Have business with bank or not	+		+
	Bank staff or not	+		
	Other loans or not	+		
	Bad credit score or not		+	
	Credit card consumption points		+	

comprehensive analysis of credit scoring, but no uniform standards exist for the indicator system of loan risk assessment. Table 1 presents the loan risk assessment indicators used by China Construction Bank (CCB), China Minsheng Bank Corp. (CMBC) and HuaXia Bank (HXB).

Work position, age, household registration, type of work, current work duration, personal income and bank account indicators are adopted by all three banks. In this paper, 11 independent variables (Table 2) are selected according to the loan evaluation indicators that are commonly used by the banks and the actual data available from the rural commercial bank. The dependent variable is a binary variable: default or no default.

3.3. Feature selection

The data have a complex distribution and substantial noise and the independent variables are composed of both continuous and discrete variables; therefore, we introduce a random forest algorithm to select the original independent variables. The most important parameters are the number of trees and attributes used to build a tree. We selected 1000 trees to constitute a forest. Then, 3 attributes were chosen for growing each decision tree; that is, 3 attributes were randomly selected from all the attributes at the split nodes of each tree as candidate characteristics. The selected variables were used to build the LRA and BPNN models based on the principal component contribution rate. The experiment in this section was performed in R (version 3.5.1) using the Ubuntu-16.04 operating system. The

principal component contribution rates of the variables are shown in Table 3.

The greater the value in Table 3 is, the greater the contribution of the variable to the final results. We selected 5 variables for the subsequent logistic regression and BPNN: loan rate, lending branch, loan amount, credit rating or not, and loan term.

3.4. Parameter tuning and setting

Generally, some parameters should be set before constructing models such as NN and stacking, and parameter tuning can be used to achieve the best training results. First, for the NN, 5 and 1 neurons were chosen for the input layer and output layer, respectively. The transfer functions between the input layer and hidden layer and between the hidden layer and output layer were set as “relu” and “sigmoid”, respectively. Meanwhile, the training method in the neural network was “adam”, a combination of the adaptive learning rate and additional momentum method. Furthermore, the learning rate was set to the default of 0.01, and the maximum number of epochs was set to 1000. A large number of studies have suggested that an appropriate number of nodes in a hidden layer is $2n + 1$, where n is the number of input nodes (Heydari, Olyaie, Mohebzadeh, & Kisi, 2013; Stathakis, 2009). Following this suggestion, we calculated the number of neurons in our hidden layer as 11. Some researchers note that a one-hidden-layer NN has a higher accuracy than a no-hidden-layer NN and a faster learning speed than a multiple-hidden-layer NN (Karsoliya, 2012; Panchal & Panchal, 2014; Vishwakarma & Gupta, 2011). Saia, Carta, and Fenu

Table 2. Variables and quantization.

Variable	Variable definitions and quantitative methods
Loan characteristics:	
Loan account	Number of loan records
Loan term	Specific days of loan
Loan amount	Principle of loan (unit: yuan)
Loan rate	Rate of loan principle
Credit guarantees	1: Hypothecated loan; 2: Mortgage loan; 3: Student loan; 4: Joint guarantee loan; 5: Peasant's micro-credit loan; 6: Credit loan; 7: Guarantee loan; 8: Housing mortgage loan
Repayment method	1: Average capital; 2: Average capital plus interest; 3: Pay interest monthly; 4: One-time payment
Lending branch	22 branches of the commercial bank
Deposit or not	1: Yes; 0: No
Credit rating or not	1: Yes; 0: No
Borrower characteristics:	
Borrower number	Customer loan number
gender	0: Female; 1: Male
age	Borrower's actual age when lending
district	A total of 86 districts

Table 3. Random forest principal component statistical results.

Independent variable	%IncMSE	IncNodePurity
Gender	12.96512	8.438386
Age	35.16297	136.364868
District	16.48660	11.312363
Loan term	92.02497	111.141352
Loan amount	92.62879	156.016574
Loan rate	245.63842	552.912804
Credit guarantees	50.17012	34.725498
Repayment ways	23.20906	5.016534
Lending branch	84.17567	137.904663
Deposit or not	30.38392	23.092034
Credit rating or not	91.42130	67.166382

(2018) performed a series of experiments to optimize the random forest performance. Inspired by their research and to avoid subjective experience, we established network models with 9-14 neurons in the hidden layer on the basis of calculating that the number of hidden layer neurons should be 11, where the number of hidden layers is varied from 1 to 3. Table 4 shows the results of the investigation of the optimal number of hidden layers and neurons: the "5-11-1" BPNN model was chosen as the final model based on the training accuracy, testing accuracy, mean square error (MSE) and learning speed.

For stacking, in order to maintain consistency with the LNN-Entropy method, two base learners (Random forest and BPNN) were chosen as the level-1 classifiers. According to the literature, simple linear models or trees with linear models at the leaves usually perform well as level-2 classifiers (Witten & Frank, 2005). Thus, in the experiments, we chose LRA as the level-2 classifier.

3.5. Performance metrics

In this paper, 4 indicators were considered to measure the prediction performance and to compare models: accuracy (ACC), area under the curve

(AUC), Kolmogorov-Smirnov statistic (KS) and Brier score (BS). We chose these metrics because they provide a comprehensive measure of model performance. ACC is a popular metric to measure the prediction power of the model. Stefan (Lessmann et al., 2013) noted that performance measures can be divided into three types: those that assess discriminatory ability, those that assess the accuracy of the models' probability predictions and those that assess the correctness of the models' categorical predictions. The AUC is a discriminatory ability metric that can accurately assess the quality of a model, even if the sample is not balanced. The KS curve charts the empirical cumulative distribution function percentages for defaults and non-defaults against the score and shows the percentage of the empirical cumulative distribution function between the default and non-default score. The KS statistic is defined by the maximum absolute difference between the two curves (Anderson, 2007), which represents whether the model has sufficient discriminatory ability for good and bad customers and can be used to assess the correctness of categorical predictions. The BS is a score function that measures the accuracy of probabilistic predictions and calculates the MSE between the predicted label (between 0 and 1) and the true label of a sample (0 or 1). The BS is defined as follows:

$$BS = \frac{1}{N} \sum_{t=1}^N (f_t - O_t)^2 \quad (6)$$

where N is the number of samples and f_t and O_t denote the predicted label and true label of sample t , respectively.

4. Experimental results

4.1. Classification results

All the experiments described in this section were performed using Python 3.6 on a server with 4 CPU

Table 4. Results of investigating the optimal number of hidden layers and neurons.

No. of layers No. of neurons	1				2				3			
	Train (%)	Test (%)	MSE	Time (s)	Train (%)	Test (%)	MSE	Time (s)	Train (%)	Test (%)	MSE	Time (s)
9	89.84	89.37	0.1063	28.96	89.87	89.45	0.1055	31.05	89.86	89.43	0.1057	33.15
10	89.85	89.42	0.1059	30.07	89.88	89.45	0.1055	32.07	89.86	89.43	0.1057	32.50
11	89.86	89.45	0.1056	28.62	89.88	89.43	0.1057	31.90	89.85	89.40	0.1057	33.87
12	89.86	89.39	0.1061	29.85	89.87	89.43	0.1057	31.42	89.85	89.40	0.1060	32.40
13	89.83	89.39	0.1062	30.55	89.86	89.45	0.1055	31.82	89.85	89.42	0.1059	33.03
14	89.85	89.34	0.1066	29.07	89.86	89.40	0.1060	33.56	89.85	89.43	0.1057	34.11

Table 5. Classifier results in terms of the different performance metrics.

Evaluation measure	Base learner		Heterogeneous	Comparison
	LRA	NN	LNN-Entropy	Stacking
ACC	0.6864	0.8976	0.9108	0.8112
AUC	0.6923	0.7137	0.7976	0.7613
KS	0.3053	0.3248	0.4839	0.3973
Brier Score	0.2135	0.0883	0.0793	0.1714

cores and 8GB memory using the Ubuntu-16.04 operating system. Our goal in this section is to show that the proposed LNN-Entropy method is competitive with the base learners and is a plausible method for credit scoring.

Table 5 summarizes the performance indicators of the base learners, LNN-Entropy method and Stacking. To minimize the influence of the variability, we repeated the experiments 100 times, applied a 100×10 -fold cross-validation and averaged the predictions.

Regarding ACC, the LNN-Entropy method improves the model performance of the base classifier LRA and NN by 22.44% and 1.32%, respectively. The AUC of LNN-Entropy is 79.76%, which indicates strong discriminatory ability between classes. Stacking places second with 76.13%. The higher the value of KS is, the better the ability to distinguish between good and bad customers. Figure 3 shows the KS values of the proposed and benchmark models: LNN-Entropy has best distinguishing ability. Finally, LNN-Entropy achieves the best BS (the lower the score is, the better): 7.93% is less than that of NN, which is in second place with 8.83%. The application of entropy and stacking substantially improves the base learners; the former provides better classification results.

There are no dominant subsets of data that influence the overall performance of the LNN-Entropy approach, since the values of the evaluation metrics remain quite stable along all the 10 subsets (i.e. the k folds used in the k -fold cross-validation process), as shown in Figure 3.

4.2. Receiver operating characteristic (ROC) curve analysis

To demonstrate the separation and discriminatory ability of the models and to evaluate their

performance and provide a measure of their sensitivity (ability to correctly classify good loans) and specificity (ability to correctly classify bad loans) over various thresholds, ROC curves are produced for the LNN-Entropy method, base classifiers and Stacking method. Figure 4 displays the ROC curves.

In an ROC curve, the true positive rate (sensitivity) is plotted as a function of the false positive rate (1-specificity) for different cut-off points. Each point on the ROC curve represents a sensitivity/specificity pair corresponding to a particular decision threshold. A test with perfect discrimination (no overlap in the two distributions) has an ROC curve that passes through the upper left corner (100% sensitivity, 100% specificity). Therefore, the closer the ROC curve is to the upper left corner, the higher the overall accuracy of the test (Zweig & Campbell, 1993). In Figure 4, the LNN-Entropy ROC curve lies above all the other curves for all threshold values; therefore, for this dataset, the LNN-Entropy method performs the best for all values of sensitivity and specificity. The Stacking ROC curve lies below that of LNN-Entropy but is above those of the other two classifiers for almost all threshold values, which means Stacking has the second-best performance.

5. Conclusion and future work

Scholars and financial institutions have made extensive contributions to credit scoring models. The aim of this study was to construct a novel model based on heterogeneous ensembles of classifiers to achieve high predictive performance. The proposed LNN-Entropy credit scoring model comprises four steps. First, data pre-processing is performed. Second, a model-based feature selection technique is applied to remove redundant variables, thereby improving the performance and decreasing the computational cost. Third, the model parameters are set; specifically, for the BPNN, previous research and experimental results are used to determine the optimal number of hidden layers and neurons. Finally, the popular logistic regression algorithm is combined with the BPNN via the entropy method to form the LNN-Entropy model, a novel heterogeneous ensemble credit scoring model, and the model performance is compared with that of two individual

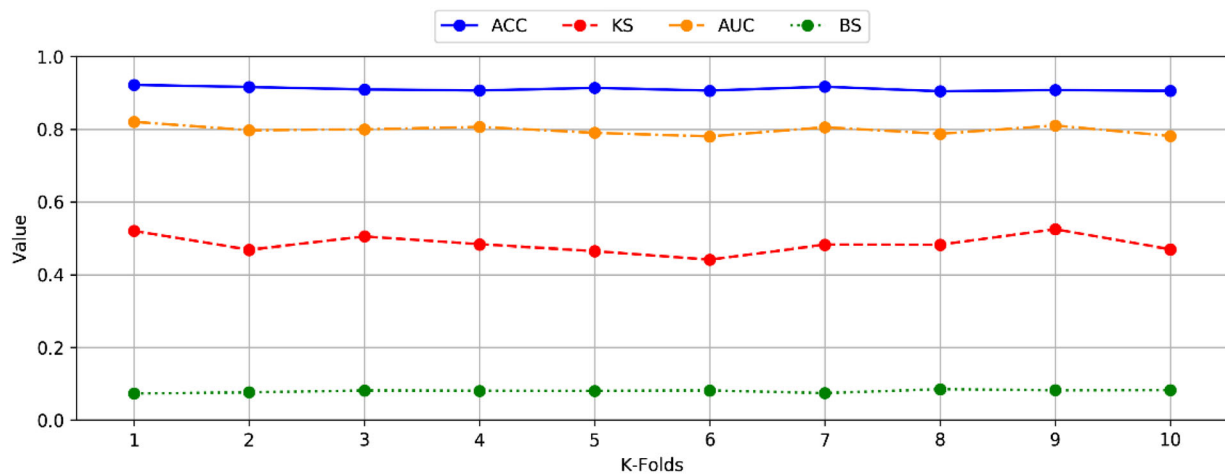


Figure 3. Performance by k -fold.

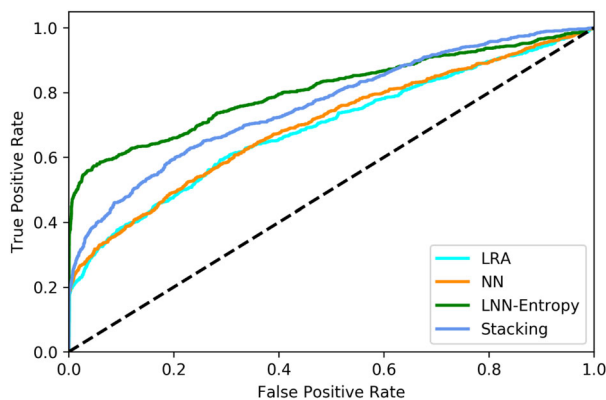


Figure 4. ROC curve comparing the performance of the LNN-Entropy approach, base classifiers and stacking.

models and a state-of-the-art model (stacking). The results show that the novel model outperforms the individual benchmark classifiers and Stacking with respect to 4 metrics: ACC, AUC, KS and BS. Even a 1% improvement in the accuracy of the credit scoring of applicants with bad credit will substantially decrease the loss for financial institutions (Hand & Henley, 1997; Wang et al., 2011). The percentage of good loans has been improved from 88.11% in the initial data to 91.08% with the LNN-Entropy model, a 2.97 percentage point increase. Improvements in prediction accuracy translate into significant future savings (Hand & Henley, 1997). The application of the novel model supplements the research on credit scoring and provides ideas for future research. With the assistance of an effective loan evaluation model, potential default loans can be driven from the market, thus invisibly improving the good-loan rate and promoting stable development of the loan market.

Some limitations of this work could be addressed in future research by (1) using more datasets of different sizes and with different attributes to validate the model, especially in the area of big data (Cao, Li, Yang, Liu, & Qu, 2018); (2) investigating other feature selection methods for the dataset; and (3)

employing other state-of-the-art classifiers, such as bagging or boosting, as baseline learners.

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